

1.

No.

DATE / /

1. 编译程序: 将一种语言编写的程序(源程序)翻译成另一种语言编写的程序(目标程序)的程序。
编译

源程序: 使用某种语言直接编写的程序。而非继续编译的源代码。

目标程序: 为用计算机, 能在目标机上执行的程序。它是已编译的源代码或汇编代码, 计算机可直接执行。

前/后端: 与用计算机/硬件部分是直接相关。

流: 从生到尾扫描一遍被编译程序。

2.

词法分析 tokenize, 分析每个词 → 单词

语法分析 parse, 根据语法规则, 分析句子的结构, 检查正误。

语义分析 检查语义错误, 如 类型不匹配。

中间代码

主, 易翻译。

codegen

效率高, 时间、空间少。

linker

→ 链接处理

→ 资源管理。

3. 边翻译边执行。区别: 无目标代码, 执行较慢, 交互性好, 适合调试。

→ 构建语法树, 进行结构分析时会发现。

4. 语法规则, 语义, 词法, 词序。

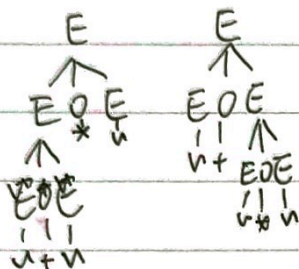
9.2.

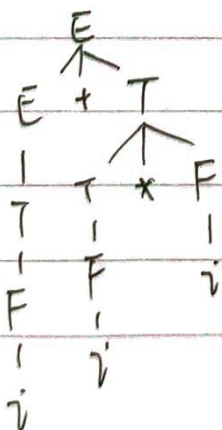
1. abc

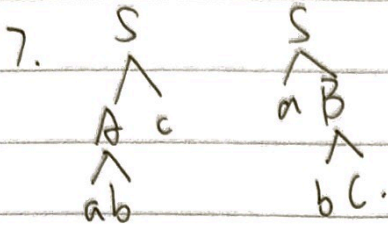
正则

 $\mathcal{N}G \rightarrow L$

2. 含任意前导0的整数

 $\mathcal{N}G \rightarrow L$ 3. $S \rightarrow A|B$ $S \rightarrow S+D|S-D|D$ $L \rightarrow G$ $A \rightarrow A+D|D$ $D \rightarrow DD'|D'$ $D' \rightarrow 1|2|\dots|9$ $D'' \rightarrow D'|0$ 4. $V+V*V$  $\mathcal{N}$ 非确定性5. $a^n b^n, n \geq 1$. $L(G[Z]) = \{a^n b^n | n \geq 1\}$ $\mathcal{N}G \rightarrow L$ 6. $G: E \rightarrow T|E+T$ $T \rightarrow F|T*F$ $F \rightarrow (E)|i$ $\mathcal{N}$ 非确定性 AST.(1) $E \Rightarrow T \Rightarrow F \Rightarrow i$ (2) $E \Rightarrow T \Rightarrow F \Rightarrow (E) \Rightarrow (T) \Rightarrow (F) \Rightarrow (i)$

$$\begin{array}{c} E \\ | \\ T \\ | \\ F \\ | \\ i \end{array}$$
(3) $E \Rightarrow T \Rightarrow T*F \Rightarrow T*i \Rightarrow F*i \Rightarrow i*i$ (4) $E \Rightarrow E+T \Rightarrow E+F \Rightarrow E+i \Rightarrow T+i$ $\Rightarrow T*F+i \Rightarrow T*i+i \Rightarrow F*i+i \Rightarrow i*i+i$ (5) $E \Rightarrow E+T \Rightarrow E+F \Rightarrow E+(E) \Rightarrow E+(E+T)$ $\Rightarrow E+(E+F) \Rightarrow E+(E+i) \Rightarrow i+(i+i)$ (6) $E \Rightarrow E+T \Rightarrow E+T*F \Rightarrow E+T*i \Rightarrow E+F*i$ $\Rightarrow E+i*i \Rightarrow T+i*i \Rightarrow F+i*i \Rightarrow i+i*i$ 

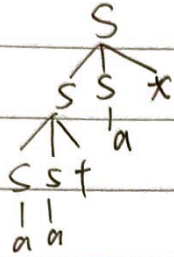


2X.

XX 判断 2X 性.

XX 推导 & AST & $G \rightarrow L$.

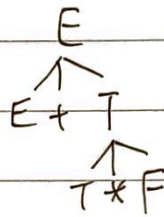
8. (1) $S \Rightarrow SS^* \Rightarrow Sa^* \Rightarrow SS+a^* \Rightarrow Sa+a^* \Rightarrow aa+a^*$



(2) 由操作符 a 包含 a^+ 组成的所有合法的
后缀表达式

9. (1) 所有合法匹配的符号串集合 (include ϵ). 戴克斯特拉 XX $G \rightarrow L$ & 2X 性.
(2) 是如 (1) (1)

10. $E \Rightarrow E+T \Rightarrow E+T^*F$



短语: $T^*F, E+T^*F$

直接: T^*F

句柄: T^*F

XX 句型 & 句柄

11. (1) $S \Rightarrow ABS \Rightarrow aBS \Rightarrow aSBBS \Rightarrow^* abbS \Rightarrow abbaa \Rightarrow abbaa$

左 $S \Rightarrow ABS \Rightarrow ABAA \Rightarrow ABaa \Rightarrow ASBBaa \Rightarrow^* abbaa$

(2) $S \rightarrow ABS | Aa | \epsilon$

$A \rightarrow a$

$B \rightarrow SBB | b$

XX 推导 & AST $\rightarrow G$ & 2X 性.

(3) 短语: $a, b, \epsilon, bb, aa, abbaa$

直接: a, b, ϵ

句柄: a

XX $L \rightarrow G$.

12. (1) $S \rightarrow aSb | ab \quad S \rightarrow aSb | \epsilon$ (4) $S \rightarrow aS | aSb | \epsilon$ ✓

(2) $S \rightarrow A \quad S \rightarrow aS | aSb | \epsilon$ (5) $S \rightarrow aSb | aSbb | aSbbb | \epsilon$

$A \rightarrow aA | aAb | \epsilon$

(6) $S \rightarrow aSa | bSb | \epsilon$

(3) $S \rightarrow WSW | a$
 $W \rightarrow a | b$

$S \rightarrow Ab$
 $A \rightarrow WAW | a$
 $W \rightarrow a | b$

$L: \{ uawb | u, w \in \{a, b\}^* \wedge |u| = |w| \}$

$$L = \{uvw^R \mid u, v, w \in \{a, b\}^+ \wedge |u| = |w| = 1\}$$

$$(7) S \rightarrow CA$$

$$A \rightarrow aAa \mid bAb \mid \underline{aCa \mid bCb} \quad X$$

$$C \rightarrow alb$$

$$(8) S \rightarrow \underline{aSa \mid bSb} \mid \underline{aa \mid bb \mid alb} \mid \epsilon$$

$$13. (1) S \rightarrow AB$$

$$A \rightarrow Aa \mid \epsilon$$

$$B \rightarrow bBc \mid \epsilon \quad \checkmark$$

$$L \rightarrow G$$

注意区别: + 与 *

大小关系

$\geq 0 \leq 1$

$$(2) S \rightarrow aSa \mid bSb \mid \epsilon \quad \checkmark$$

$$\epsilon \rightarrow alb$$

$$(3) S \rightarrow AB \mid A'B'$$

$$A \rightarrow A$$

$$A \rightarrow aA \mid \epsilon$$

$$B \rightarrow bBc \mid \epsilon$$

$$C \rightarrow aCb \mid \epsilon$$

$$D \rightarrow cD \mid \epsilon \quad \checkmark$$

或"或"分类讨论, $S \rightarrow S_1 \mid S_2$

与两题讨论

$$(4) S \rightarrow S' \mid AB$$

$$S' \rightarrow aS'c \mid S''$$

$$S'' \rightarrow bS'' \mid \epsilon \quad \checkmark$$

hard

$$(5) 1 \leq k \leq n \leq k+m$$

$$S \Rightarrow \frac{1}{2} n = k+x \quad (x \geq 0) \quad \text{由 } n \leq k+m$$

$$\frac{1}{2} m = 1+y \quad (y \geq 0)$$

$$\Rightarrow n \leq m$$

$$\Rightarrow T(n) = a^{k+x} b^{1+y} c^k = a^k \cdot a^x \cdot b^1 \cdot b^y \cdot c^k$$

$$S \rightarrow aSc \mid aAc \quad (k \geq 1)$$

$$A \rightarrow aAb \mid aBb \mid bB \quad (m \geq 1)$$

$$B \rightarrow bB \mid \epsilon \quad S \rightarrow S_1 \mid S_2$$

$$(6) S \rightarrow aSc \mid aS \mid B$$

$$B \rightarrow bB \mid \epsilon \quad \checkmark$$

$$(7) S \rightarrow abScd \mid abcd$$

$$S \rightarrow aSd \mid B$$

$$B \rightarrow bcB \mid bc$$

$$S \rightarrow A \mid B$$

$$A \rightarrow aA \mid abA \mid cd \mid abcd$$

$$B \rightarrow aBd \mid aCd$$

$$C \rightarrow bc \mid bc$$

$$bCc$$

$$A \rightarrow aAb \mid ab$$

$$B \rightarrow cBd \mid cd$$

$$S_2 \rightarrow aS_2d \mid aCd$$

$$C \rightarrow bCc \mid bc$$

WCWC^R

⑧ $S \rightarrow W'B$ $W' = \cancel{W} W' Wc | \epsilon$
 $W \rightarrow aW | bW | a | b$
 $B \rightarrow \cancel{a} \cancel{b} \cancel{c} \cancel{a} aBa | bBb | \cancel{W} \cancel{Wc} \cancel{a} \cancel{W'a} | bW'b$
 整理: $\Rightarrow S \rightarrow LM$
 $L \rightarrow WcL | \epsilon$
 $M \rightarrow aMa | bMb | \underline{aXa | bXb}$
 $X \rightarrow cLc$
 同 12. (7).

14. $S \rightarrow \$ (S) S | \$ [S] S | \epsilon$ $\cancel{L \rightarrow G}$

15. (1) $S \rightarrow DD_2$ (2) $S \rightarrow DD_2$
 $D \rightarrow D_0 D | \epsilon$ $D \rightarrow D_1 D'_1 | \epsilon$ $\cancel{L \rightarrow G}$
 $D'_1 \rightarrow D_1 | 0$ $\cancel{D \rightarrow aEb | bEa | EE | \epsilon}$

⑩ $S \rightarrow S_1 | S_2 | S_3$ $\cancel{L \rightarrow G}$
 $S_1 \rightarrow \cancel{a} \cancel{c} \cancel{S_1} \cancel{c} \cancel{a}$ $\cancel{S_2 \rightarrow aS_2b | bS_2a}$
 $\begin{matrix} \uparrow & \uparrow & \uparrow \\ a & b & b^*a \end{matrix}$
 $\cancel{S_2 \rightarrow aS_2b | bS_2a}$ $\cancel{S_3 \rightarrow S_a | S_b}$
 $S_a \rightarrow EaS_a | EaE$
 $S_b \rightarrow EbS_b | EbE$

17. 不是. CFG 仅能处理 嵌套式 语言, 而不能处理 交叉式 语言. $\cancel{CFG}$
 特性: 按序匹配左右匹配.

18. (1) $S \rightarrow aS | \epsilon$ $\cancel{L \rightarrow G}$ & 3 型文法.
 (2) $S \rightarrow aA$ $S \rightarrow aS | aB$ (3) $S \rightarrow aA$ $C \rightarrow cC | \epsilon$ $S \rightarrow aAB$
 $A \rightarrow aA | bB$ $B \rightarrow bB | b$ $A \rightarrow aA | bB$ $B \rightarrow bB | cC$ $C \rightarrow cC | \epsilon$
 $B \rightarrow bB | \epsilon$ $C \rightarrow cC | \epsilon$ $A \rightarrow B$ $\cancel{A \rightarrow B}$ 3 型文法
 $\epsilon \notin V_T$

$\Lambda = \lambda$

19. (1) if () < > if () < > else < >

 $\Rightarrow = \lambda$

②

仍能
生成

(2) key: else 究竟归谁?

↓

else 前的 if 中. 所有 if 必须有 else 匹配,

why 保持了语义? 此处仅考虑 $L(G < stmt >)$ 是否相同. 不能引大括号.

20. 找语义错误.

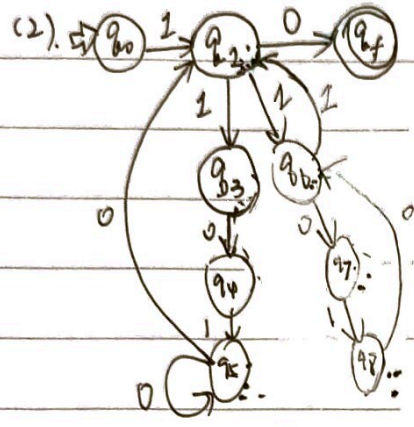
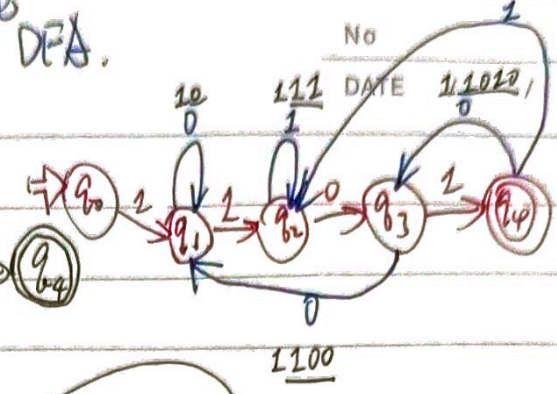
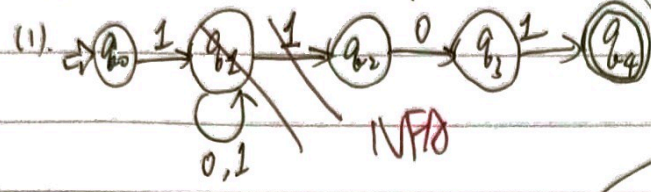
书也能办物理了. but we don't care. we only care about the final expression.

Λ 语义. λ

~~NFA~~ $\rightarrow$ DFA

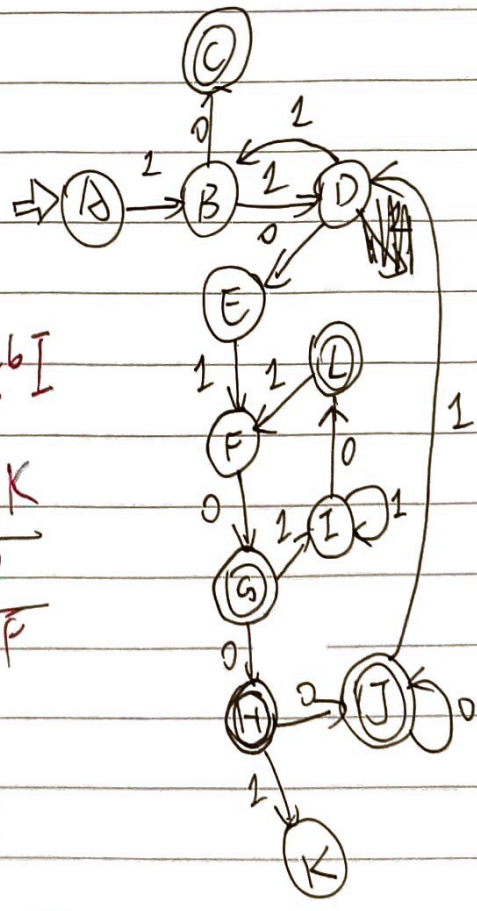
Q.3.

1. $\{K, \Sigma, f, S, Z\}$

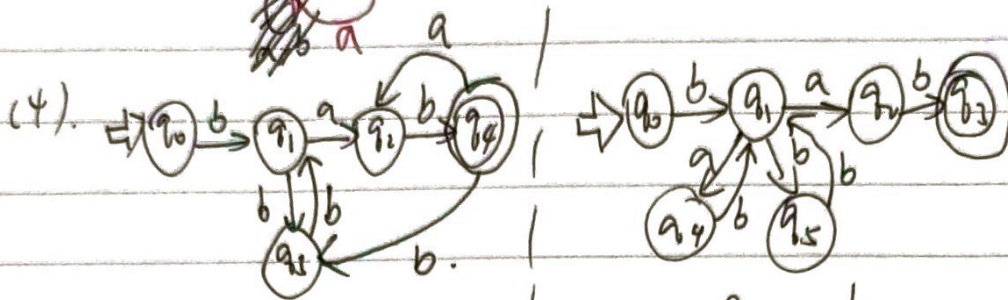
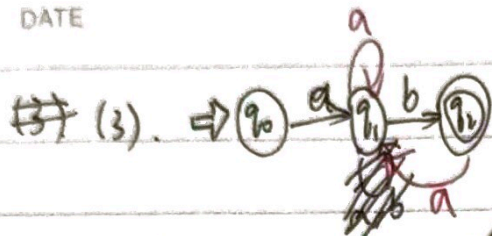


$\{0/0^* \mid (0/0)^* \mid 0\}$

0.A $\emptyset$ 2B
 2B fC 3,6D
 fC $\emptyset$ $\emptyset$
 3,6D 4,7E 2B
 4,7E $\emptyset$ 5,8F
 5,8F 2,5,6G $\emptyset$ 2,3,6I
 2,5,6G 2,5,7,fH ~~3,6I~~
 2,5,7,fH 2,5,fJ 3,6,8K
~~3,6,8I~~ ~~4,6,7J~~ 2B
~~4,6,7J~~ 7K ~~5,8F~~
~~7K~~ $\emptyset$ ~~8L~~
~~8L~~ 6M $\emptyset$
~~6M~~ 7K 2B



2, 3,6 I 4,7,f L 2,3,6 I
 2,5,f J 2,5,f J 3,6D 4,7,f L $\emptyset$ 5,8F
 3,6,8 K 4,6,7 ~~2B~~ 4,6,7 7 2,5,8
~~4,6,7 L~~ ~~7 2,5,8~~



~~非~~ NFA 公式 转换图

总结: 由正规式求 DFA.

正规式化简

$\downarrow$
NFA 转换图

$\downarrow$
子集法

$\downarrow$
DFA 转换图

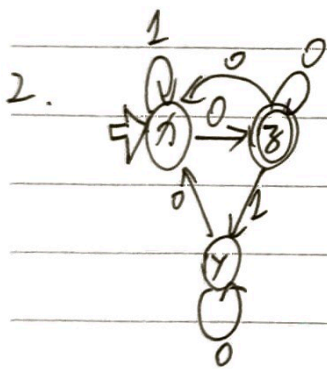
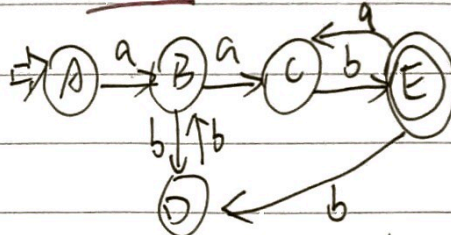
$\begin{matrix} & a & b \\ DA & 1B & \phi \end{matrix}$

$\begin{matrix} 1B & 2,4C & 5D \end{matrix}$

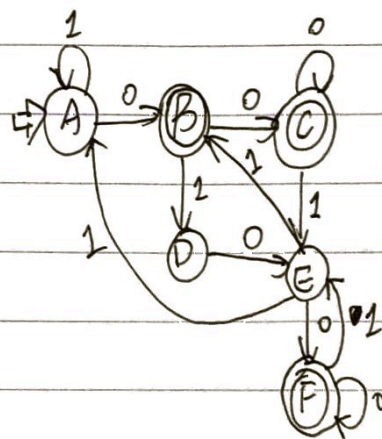
$\begin{matrix} 2,4C & \phi & 1,3E \end{matrix}$

$\begin{matrix} 5D & \phi & 1B \end{matrix}$

$\begin{matrix} 1,3E & 2,4C & 5D \end{matrix}$



$\begin{matrix} & 0 & 1 \\ \rightarrow A & 2B & 1A \\ 2B & 1,2C & 1D \\ 1,2C & 1,2C & 1E \\ 1D & 1,2E & 2B \\ 1,2E & 1,2E & 1A \\ 1,2E & 1,2E & 1,2E \end{matrix}$

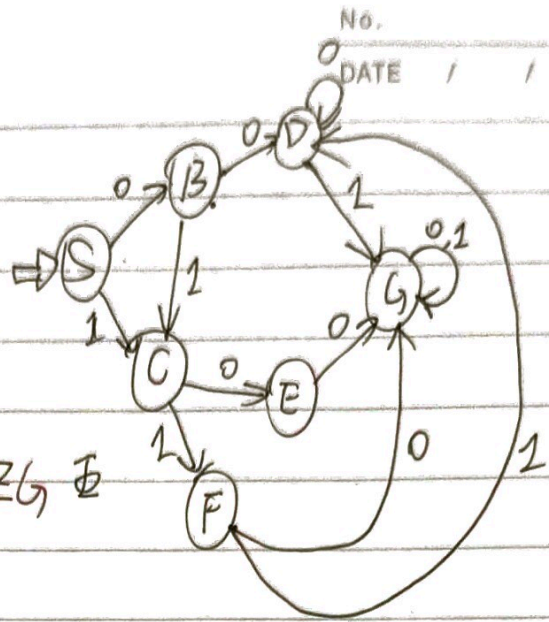


~~NFA $\Rightarrow$ DFA~~

~~NFA~~ → DFA

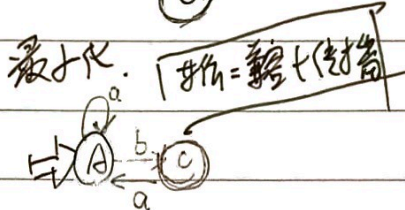
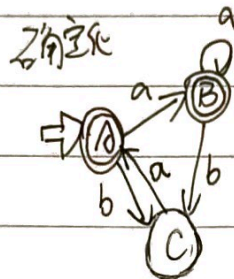
3.

$\rightarrow SA \quad VQB \quad QUC$
 $VQB \quad VZD \quad QUC$
 $QUC \quad V.E \quad QZF$
 $VZD \quad VZD \quad ZG$
 $QZF \quad ZG \quad VZD$
 $ZG \quad ZG \quad ZG$

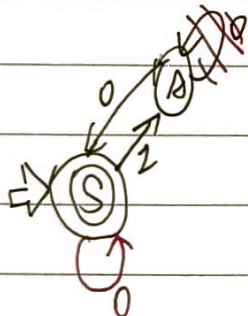


4.

$\rightarrow 0A \quad 01B \quad 1C$
 $01B \quad 01B \quad 1C$
 $1C \quad 0A$



5.



$S \rightarrow 0S \mid 1A \mid \epsilon$
 $S \rightarrow 0S \mid 1A \mid \epsilon$
 $A \rightarrow 0S$

~~L → DFA → RG.~~

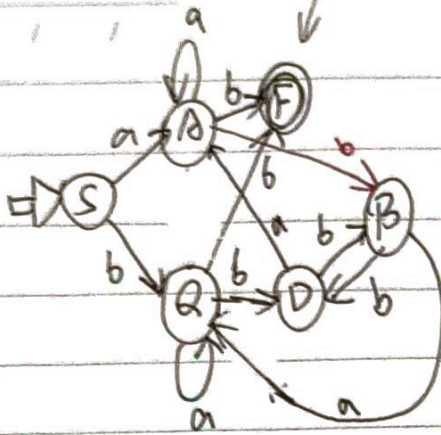
6. $\theta = dd^* (\epsilon \mid dd^* (\epsilon \mid 10^A dd^*))$

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引入新状态.

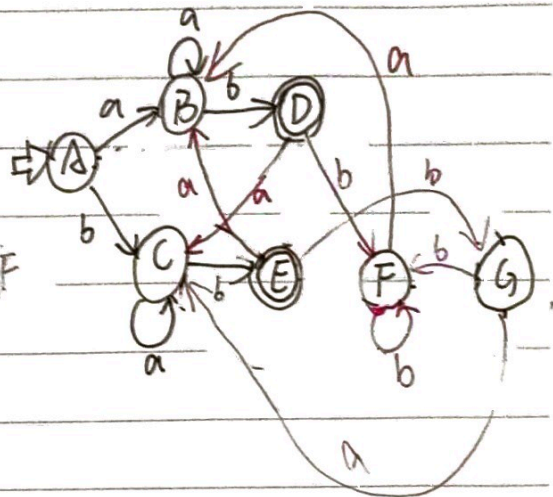
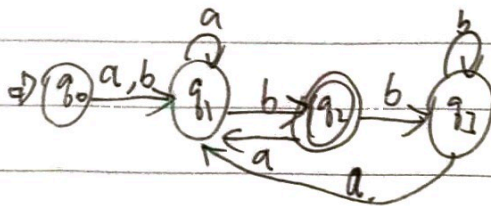
7.



E, F 无用. 删去.

 $\Rightarrow$ 分析 $\Rightarrow$ 化简.

$\rightarrow$ S	A	B	C
A	B	B	BE
QC	QC	DFE	
BE	QC	DF	
DF	E	AB	BG
DF	AB	BF	
BG	AC	DF	

 $\Pi_0: F = \{D, E\}, N = \{A, B, C, F, G\}$ a/b 均 $\in$ 非终态
& 均 $\in N$ a/b $\{A, F, G\} \quad \{B, C\} \quad \Pi_1$ $\{A\}, \{F, G\} \quad \Pi_2$ a/b $\pi_0 = \{A\}, \pi_1 = \{B, C\}, \pi_2 = \{D, E\}, \pi_3 = \{F, G\}$ RG $\rightarrow$ NFA
 $\downarrow$ 构造
最小
DFA Δ 化 = 删除无用状态 + 合并等价状态. Δ RG $\rightarrow$ NFA: 构造序列为 (2).

8. 方程代入法

$$A = 1S + 1 = 1(S + \epsilon)$$

$$B = 0S + 0 = 0(S + \epsilon)$$

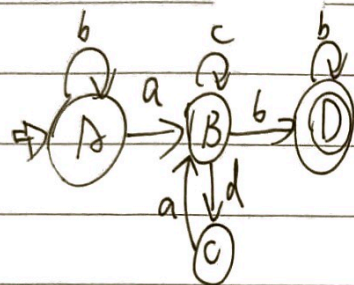
~~RG~~ $\Rightarrow RE$

$$S = 0(1(S + \epsilon)) + 1(0(S + \epsilon))$$

$$= (01 + 10)(S + \epsilon)$$

$$= (01 + 10)S + (01 + 10)$$

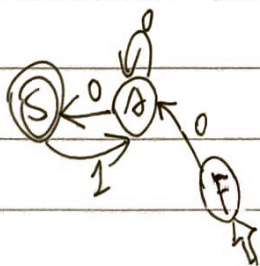
$$\Rightarrow S = (01 + 10)^*(01 + 10) = \underline{\underline{(01 + 10)^+}}$$

9. $\Pi_1: F = \{6, 7\}, N = \{1, 2, 3, 4\}$ 

$\{1, 2, 5\}, \{3, 4\}$
 $\{2, 2\}, \{5\}$
 $A \quad C$

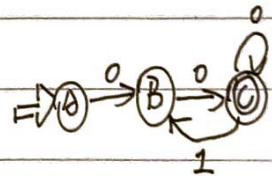
~~正则化~~
 $DFA \rightarrow RE$
 $\Rightarrow$ 正规式: $b^*a(ad|c)^*b^*b$

10.



不确定构造

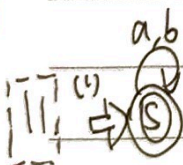
$\rightarrow F$ A AB 重
 $\rightarrow B$ AB ASC 重
 $\rightarrow C$ ASC AB



注意:

正规式: $0(0+1)^*0$ $00(0+1)^*$ $\Rightarrow$ 00 开头, each 1

has 0 after 1

all 0, 1 组成的
字符串.(2) $\Rightarrow S$

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13. $\text{①} \rightarrow \text{②}$
~~最长匹配原则~~ + ~~左优先原则~~
匹配更长的串 排在后面的规则优先.
lex.

14. abbacbabcb

5 abb

4 ac

3 baba

2 bc

1 ab.

Q. 4.

FIRST: 含 ϵ .
FOLLOW: 不含 ϵ .
SELECT: 不含 ϵ .

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LL(1) 分析

1. (1) $S \Rightarrow (T) \Rightarrow (T, S) \Rightarrow (S, S) \Rightarrow (a, S) \Rightarrow (a, (T)) \Rightarrow (a, (T, S))$
 $\Rightarrow (a, (S, S)) \Rightarrow (a, (a, S)) \Rightarrow (a, (a, a))$

10分

(2) $S \rightarrow a \mid \wedge \mid (T)$
 $T \rightarrow ST'$
 $T' \rightarrow , ST' \mid \epsilon$

~~$T \rightarrow ST'$~~ First 冲突

Code: void S() {

```
if (sym == 'a') {
    advance();
} else if (sym == '\wedge') {
    advance();
} else if (sym == '(') {
    advance();
    T();
    if (sym == ',') advance();
    else error();
} else {
    error();
}
```

void T() {

```
S();
T_prime();
}

void T_prime() {
    if (sym == ',') {
        advance();
        S();
        T_prime();
    }
}
```

~~(3)~~

FIRST(S) = {a, $\wedge$, (}
T {a, $\wedge$, (}
T' {, , ϵ }

FOLLOW(S) = {, ,), #}
T {)}
T' {)}

FOLLOW(S) = {#} $\cup$ FIRST(T') - { ϵ }

FOLLOW(T') = FOLLOW(T') $\cup$ FOLLOW(T)

~~SELECT(S $\rightarrow$ a | $\wedge$ | (T)) = {~~

SELECT(S $\rightarrow$ a) = {a}
S $\rightarrow$ $\wedge$ { $\wedge$ }
S $\rightarrow$ (A) {(}
T $\rightarrow$ ST' {a, $\wedge$, (}
T' $\rightarrow$, ST' {,}
T' $\rightarrow$ ϵ {)}

冲突

$\Rightarrow$ 符合 LL(1).

	a	$\wedge$	()	,
S	S $\rightarrow$ a	S $\rightarrow$ $\wedge$	S $\rightarrow$ (A)	
T	T $\rightarrow$ ST'	T $\rightarrow$ ST'	T $\rightarrow$ ST'	
T'			T $\rightarrow$ ϵ	T $\rightarrow$ ST'

(4).	步骤	符号栈	输入串	动作
1	#S	(a,a)#	查表 $S \rightarrow (T)$	
2	#)T(	(a,a)#	匹配(	
3	#)T	a,a)#	查表 $T \rightarrow ST'$	
4	#)T'S	a,a)#	查表 $S \rightarrow a$	
5	#)T'a	a,a)#	匹配a	
6	#)T'	,a)#	查表 $T' \rightarrow ,S$	
7	#)T'S,	,a)#	匹配,	
8	#)T'S	a)#	查表 $S \rightarrow a$	
9	#)T'a	a)#	匹配a	
10	#)T'	)#	查表 $T' \rightarrow \varepsilon$	
11	#)	)#	匹配)	
12	#	#	接受	

2. (1). $FIRST(E) = \{ (, a, b, \wedge \}$ $FOLLOW(E) = \{ \#,) \}$

$E' \rightarrow \{ +, \varepsilon \}$

$T \rightarrow \{ (, a, b, \wedge \}$

$T' \rightarrow \{ (, a, b, \wedge, \varepsilon \}$

$F \rightarrow \{ (, a, b, \wedge \}$

$F' \rightarrow \{ *, \varepsilon \}$

$P \rightarrow \{ (, a, b, \wedge \}$

$E' \rightarrow \{ \#,) \}$

$T \rightarrow \{ +,), \# \}$

$T' \rightarrow \{ +,), \# \}$

$F \rightarrow \{ +,), \# \}$

$F' \rightarrow \{ +,), \# \}$

$P \rightarrow \{ +,), \# \}$

(2). $SELECT(E \rightarrow TE) = \{ (, a, b, \wedge \}$

$(E' \rightarrow TE) = \{ + \}$

$E' \rightarrow \varepsilon \rightarrow \{), \# \}$

$T \rightarrow T \rightarrow \{ (, a, b, \wedge \}$

$T \rightarrow \varepsilon \rightarrow \{ +,), \# \}$

$F \rightarrow PF \rightarrow \{ (, a, b, \wedge \}$

$F' \rightarrow KF \rightarrow \{ * \}$

$F' \rightarrow \varepsilon \rightarrow \{ (, a, b, \wedge, +,), \# \}$

$FOLLOW(P) = FIRST(E') \setminus \{ \varepsilon \} \cup FOLLOW(E)$

$P \rightarrow (E) = \{ (\}$

$P \rightarrow a \rightarrow a$

$P \rightarrow b \rightarrow b$

$P \rightarrow \wedge \rightarrow \wedge$

$\Rightarrow$ 符合 LL(1).

(3) $E \rightarrow TE' \quad E \rightarrow TE' \quad E \rightarrow TE' \quad E \rightarrow TE'$ E' $E' \rightarrow TE \quad E' \rightarrow \epsilon \quad E' \rightarrow \epsilon$ $T \rightarrow FT' \quad T \rightarrow FT' \quad T \rightarrow FT' \quad T \rightarrow FT'$ $T' \rightarrow T \quad T' \rightarrow \epsilon$ $F \rightarrow PF'$ $F' \rightarrow \epsilon$ $P \rightarrow (E) \quad P \rightarrow a \quad P \rightarrow b \quad P \rightarrow \wedge$

(4) 略.

*LL(1).

(3) $FIRST(S) = \{d, e, b, a\}$ $FIRST(S) = (FIRST(M) / \{\epsilon\})$ $H \rightarrow \{e, \epsilon\}$ $U(FIRST(H) / \{\epsilon\})$ $K \rightarrow \{d, \epsilon\}$ $U\{a\}$ $L \rightarrow \{e\}$ $U\{\epsilon\}$ $M \rightarrow \{d, e, b\}$ $= \{a, b, d, e, \epsilon\}$ $FOLLOW(S) = \{o, \# \}$ $H \rightarrow \{t, o, \# \}$ $K \rightarrow \{e, o, \# \}$ $L \rightarrow \{a, b, d, e, o, \# \}$ $FOLLOW(L) = (FIRST(S) / \{\epsilon\})$ $M \rightarrow \{e, o, \# \}$ $U(FOLLOW(S))$ $SELECT(S \rightarrow MH) = \{d, b, e, o, \# \}$ $U(FIRST(M) / \{\epsilon\})$ $S \rightarrow a \quad \{a\}$ $U(FOLLOW(M))$ $H \rightarrow LSo \quad \text{略}.$ $U(FOLLOW(K))$ $H \rightarrow \epsilon$ $K \rightarrow dML$ $K \rightarrow \epsilon$ $L \rightarrow eHf$ $M \rightarrow K$ $M \rightarrow bLM$

4. (4) ~~FIRST(a) = {a}~~ $\rightarrow$ 冲突 $\rightarrow$ ϵ LL(1)
~~(a) {a}~~

~~LL(1)~~
 改3

(2). 不能. 提取左因子, 仍会冲突.
 略.

5. key: ① $L \rightarrow S | L; S \Rightarrow L \rightarrow S L'$

$L' \rightarrow \epsilon | L'$

~~LL(1)~~
 改3.

② $C \rightarrow I | I \text{ else } S \Rightarrow C \rightarrow I C'$

略.

$C' \rightarrow \epsilon | \text{else } S$

6. (1) 提取左因子.

(2) 提取左因子

(3). $\checkmark$

(4). 左因子

(5). 左因子

(6). 左因子+左因子. $M \rightarrow H M'$

$M' \rightarrow a H M' | \epsilon$

$H \rightarrow b B | (M)$

$B \rightarrow (M) | \epsilon$

~~LL(1)~~
 改3.

7. (1). $A \rightarrow \cancel{ba}bb | \cancel{ba}a | \epsilon \Rightarrow ba B | \epsilon$

$B \rightarrow \cancel{A}b | \cancel{ba}A | \epsilon \Rightarrow Ab | a$

$A \Rightarrow B \Rightarrow B \rightarrow (ba B | \epsilon) bb | a$

$= ba B bb | bb | a$

$= b B' | a$

$B' \rightarrow a B bb | b$

~~LL(1)~~
 改3.

$$(2). A \rightarrow a A'$$

$$A' \rightarrow AB\epsilon | \epsilon$$

$$B \rightarrow dB'$$

$$B' \rightarrow bB' | \epsilon$$

✓

$$(3). S \rightarrow Saba | b$$

$$\Rightarrow S \rightarrow bS'$$

$$S' \rightarrow abaS' | \epsilon$$

✓

$$(4). \cancel{S \rightarrow SA} \quad S$$

$$A \rightarrow ASA | bA | a$$

$$A \rightarrow aA'$$

$$A' \rightarrow S$$

$$\Rightarrow A \rightarrow bAA' | aA'$$

$$A' \rightarrow SAA' | \epsilon$$

$$+ S \rightarrow AS | b \quad \text{FIRST}(AS) = \{a, b\} \cap \text{FIRST}(b) = \{b\} \neq \emptyset$$

X.

$$(5). \cancel{S \rightarrow SA} \quad A \rightarrow aA' \quad A \rightarrow aA'$$

$$A' \rightarrow aA' | \epsilon$$

X

(6)

X

$$8. \quad \cancel{S = PQ + a} \quad - \text{S.R.P.}$$

$$\cancel{P = QS + b}$$

$$\cancel{Q = SP + c}$$

$$S \rightarrow PQ | a$$

$$Q \rightarrow SP | c = P \& P | aP | c$$

$$P \rightarrow QSh = P \& PS | aPS | cS | b$$

$$\Rightarrow P \rightarrow aPS' | cS' | bP'$$

$$P' \rightarrow QPS' | \epsilon$$

7/2/20

9. 递归归

LL(2)

first + follow

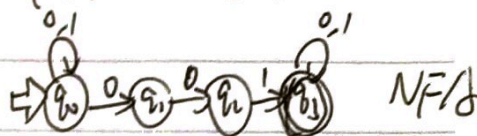
select.

构造预测表

递归归

10. $S \rightarrow aSbS \mid bSas \mid \epsilon$ LL(1)

LL(2)

~~LL(1)~~ $S \rightarrow aAS \mid bBS \mid \epsilon$ $A \rightarrow aA \mid b$ $B \rightarrow bB \mid a$ 1. $\{a^m b^n c^k \mid m, n, k \geq 0, m \geq k\}$ 2. $S \rightarrow S+S \mid S^* \mid a$ ~~LL(1)~~3. $(01)^* 001 (01)^*$ $m = k + p \ (p \geq 0)$ $a^k a^p b^n c^k$ $S \rightarrow AX$ $A \rightarrow aA \mid \epsilon$ $X \rightarrow aXc \mid Y$ $Y \rightarrow bY \mid \epsilon$ 

NFA

↓
正则↓
正则↓
DFA4. $B \rightarrow A$. $A \rightarrow (A|a) \mid a|b$ $= Aca \mid a|b$

$$\begin{cases} A \rightarrow \cancel{a} A' \mid bA' \\ A' \rightarrow caA' \mid \epsilon \\ C \rightarrow ab \end{cases}$$

例6.

1. $FOLLOW(A) = \{b, d, \#\}$

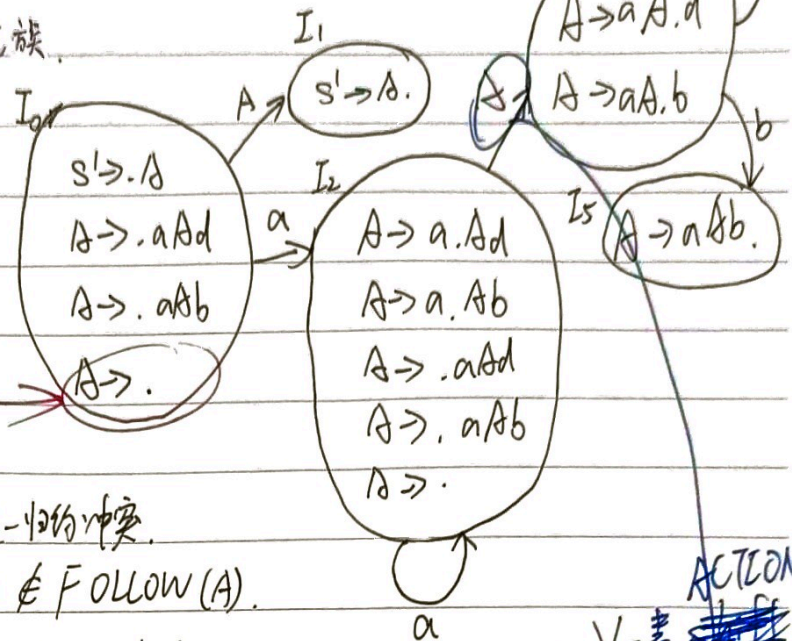
②构造LR(0)项目集规范族.

G: (0) $S' \rightarrow A$

(1) $A \rightarrow aAd$

(2) $A \rightarrow aAb$

(3) $A \rightarrow \epsilon$



③ I_0, I_2 均存在移进-归约冲突.

遇 a 移进, 但 $a \notin FOLLOW(A)$.

$\therefore$ 该文法可用 SLR(1) 方法消峰.

故该文法为 SLR(1) 文法.

④ 构造分析表.

状态	ACTION				GOTO
	a	b	d	#	
0	S2	r3	r3	r3	1
1					
2	S2	r3	r3	r3	3
3		S5	S4		
4		r1	r1	r1	
5		r2	r2	r2	

$FOLLOW(A)$ 中任一符号, 皆可归约.

归约可否, 取决于 $FOLLOW(A)$ 中是否有.

⑤ 分析:

步骤	状态栈	符号栈	输入串	ACTION	GOTO
1	0	#	ab#	S2	
2	02	#a	b#	r3	3
3	023	#aA	b#	S5	
4	0235	#aAb	#	r2	
5	01	#A	#	acc	

1. 归约时, 要 GOTO.
归约会消掉几个
符号及状态, 根
据状态栈顶的
状态查 GOTO 表.

2. 遇到符号, 归约
归约时, 符号栈会消几个, 状态栈同时 pop 几个.
归约时, 符号栈会消几个, 状态栈同时 pop 几个.

No

DATE / /

(1) (10) $S' \rightarrow S$
2. G: (11) $S \rightarrow L.L$

$FOLLOW(S) = \{\#\}$

(12) $S \rightarrow L$

$FOLLOW(L) = \{., 0, 1, \#\}$

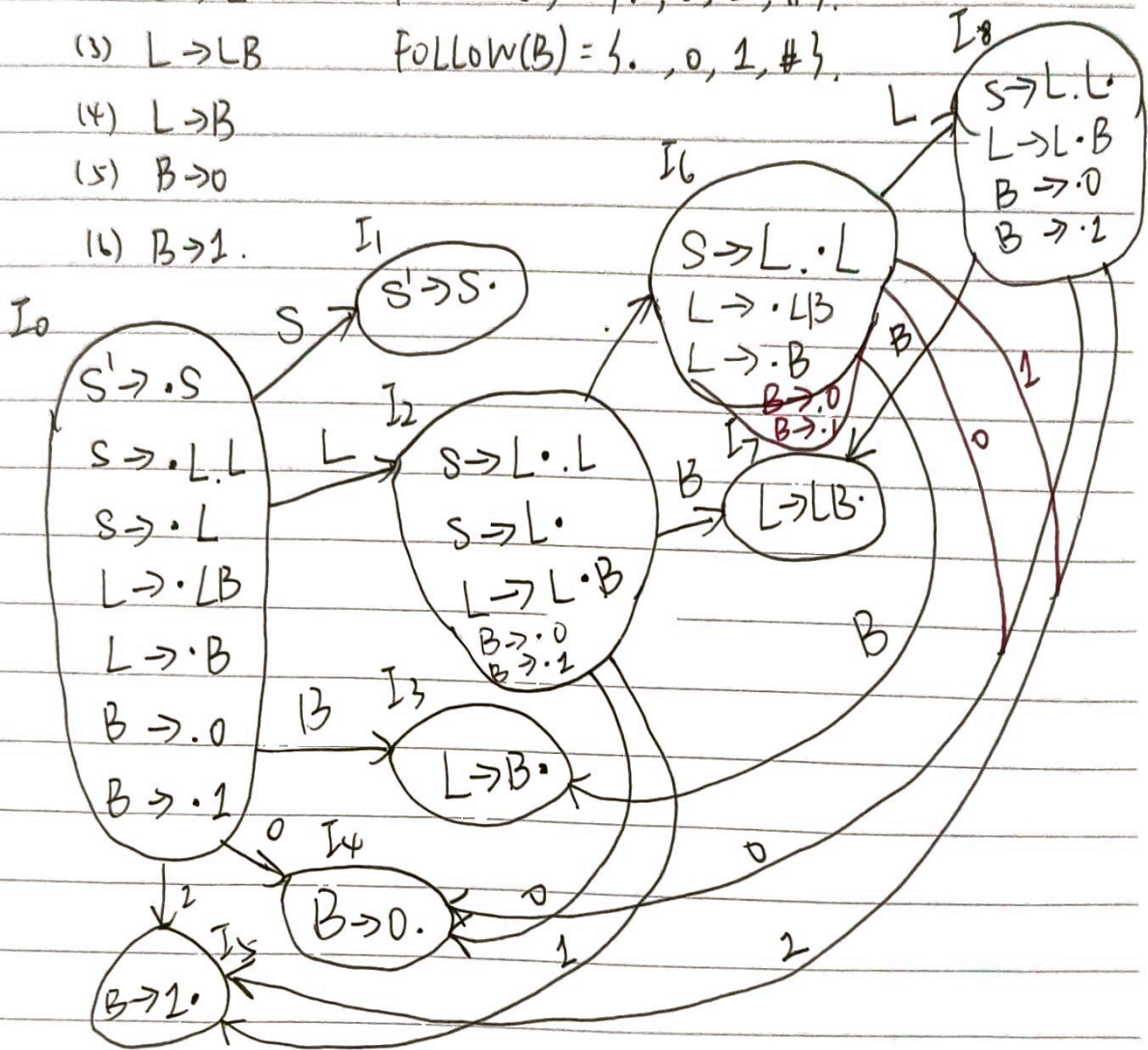
(13) $L \rightarrow LB$

$FOLLOW(B) = \{., 0, 1, \#\}$

(14) $L \rightarrow B$

(15) $B \rightarrow 0$

(16) $B \rightarrow 1$



I_2 : 移进 $\{., 0, 1\}$ 与归约 ($S \rightarrow L$) 冲突. $FOLLOW(S) = \{\#\} \cap \{0, 1, .\} = \emptyset$
= 冲突可消解.

I_7 : 移进 $\{0, 1\}$ 与归约 ($S \rightarrow L.L$) 冲突. $FOLLOW(S) = \{\#\} \cap \{0, 1\} = \emptyset$
= 冲突可消解.



SLR(1) 分析表.

状态	Action				Goto.		
	0	1	.	#	S	L	B
0	S4	S5			1	2	3
1				acc			
2	S4	S5	S6	r2			7
3	r4	r4	r4	r4			
4	r5	r5	r5	r5			
5	r6	r6	r6	r6			
6	S4	S5				8	3
7	r3	r3	r3	r3			
8	S4	S5		r1			7

(2).

状态 状态码 符号串 操作 Action Goto.

1	0	#	101.110#	S8	
2	05	#1	01.110#	r6	3
3	03	#B	01.110#	r4	2
4	02	#L	01.110#	S4	
5	024	#L0	1.110#	r5	7
6	027	#LB	1.110#	r3	2
7	02	#L	1.110#	S5	
8	025	#L1	.110#	r6	7
9	027	#LB	.110#	r3	2
10	02	#L	.110#	S6	
11	026	#L	110#	S4	

.....

20	0268	#L.L	#	r1	1
21	01	#S	#	acc	

3. (1) $S' \rightarrow .S$ $S' \rightarrow S.$ $S \rightarrow .AS$ $S \rightarrow A.S$ $S \rightarrow AS.$

.....

~~LR(0)~~ & 活前缀
& 1/4

(2).

4. ~~LALR(1)~~ / SLR(1).5. ~~LR(0)~~.~~10.~~

(1) SLR(1).

(2) CLR . 2义

(3) LR(1).

(4) CLR . S-冲突

(5) CLR . 非确定归约时机

(6) LR(0).

11. (11). GCS]: $S' \rightarrow S$

1. $S \rightarrow AS$

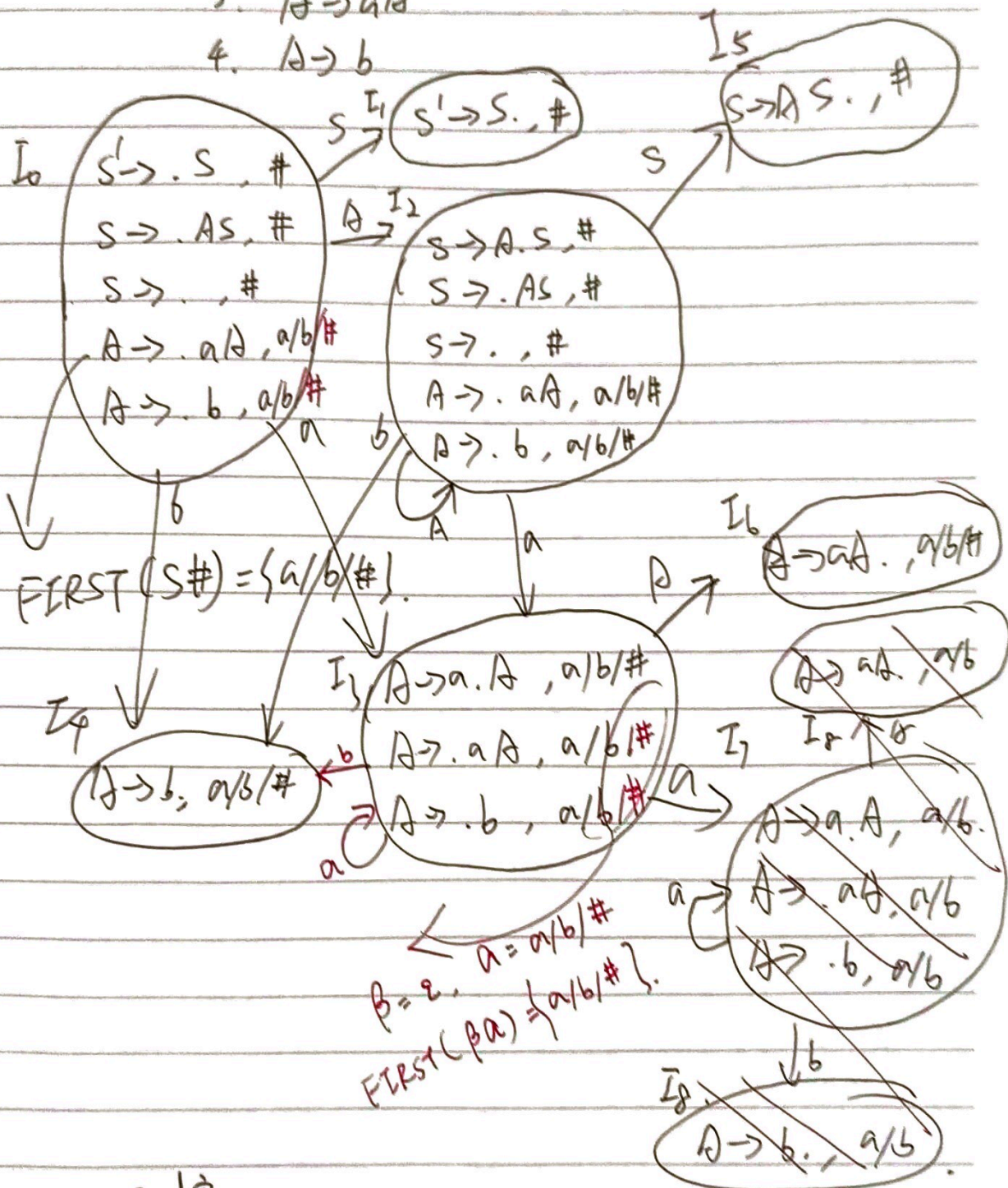
2. $S \rightarrow \epsilon$

3. $A \rightarrow aA$

4. $A \rightarrow b$

~~$FIRST(S) = \{a, b, \epsilon\}$~~

~~$\{a, b\}$~~



∴ B 冲完

∴ GCS] 的 LR(1) 文法.

(3)	栈	栈顶	栈底	ACTION	输出
1	0	#	abab#	S ₃	
2	03	#a	bab#	S ₄	
3	034	#ab	ab#	r ₄	6
4	036	#aA	a.b#	r ₃	2
5	02	#A	ab#	S ₃	
6	023	#Aa	b#	S ₄	
7	0234	#Aab	#	r ₄	6
8	0236	#AaA	#	r ₃	2
9	022	#AA	#	r ₂	5
10	0225	#AAS	#	r ₁	5
11	025	#AS	#	r ₁	1
12	01	#S	#	all	

→ 2)

No	ACTION			Cost	
	a	b	#	A	S
0	S ₃	S ₄	r ₂	2	1
1			all		
2	S ₃	S ₄	r ₂	2	5
3	S ₃	S ₄		6	
4	r ₄	r ₄	r ₄		
5			r ₁		
6	r ₃	r ₃	r ₃		

No

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1. $S \rightarrow AS$

2. $S \rightarrow \epsilon$

3. $A \rightarrow a/b$

4. $A \rightarrow b$

(18) (1) ~~$I_0: S \rightarrow S$~~ (2) $follow(S) = \{e, \#\}$. $A \rightarrow e/\#$

~~$I_2: S \rightarrow S$~~

~~$I_4: S \rightarrow r$~~

~~$I_5: S \rightarrow S$~~

$S \rightarrow S$

$S \rightarrow iSeS$

$S \rightarrow iS$

$S \rightarrow a$

(3)

$I_0:$

$S \rightarrow .S, \#$
 $S \rightarrow .iSeS, \#$
 $S \rightarrow .iS, \#$
 $S \rightarrow .a, \#$

$S \rightarrow S, \#$

$I_4:$

$S \rightarrow iS.eS, \#$
 $S \rightarrow iS, \#$

$I_1 \downarrow a$

$S \rightarrow a, \#$

$S \rightarrow i.SeS, \#$
 $S \rightarrow i.S, \#$
 $S \rightarrow .iSeS, e/\#$
 $S \rightarrow .iS, e/\#$
 $S \rightarrow .a, e/\#$

$I_7:$

$S \rightarrow iSe.S, \#$
 $S \rightarrow .iSeS, \#$
 $S \rightarrow .iS, \#$
 $S \rightarrow .a, \#$

$I_5:$

$S \rightarrow a., e/\#$

$I_5 \downarrow i$

$S \rightarrow i.SeS, e/\#$
 $S \rightarrow i.S, e/\#$
 $S \rightarrow .iSeS, e/\#$
 $S \rightarrow .iS, e/\#$
 $S \rightarrow .a, e/\#$

~~I_4~~

~~$S \rightarrow iS.eS, e/\#$~~
 ~~$S \rightarrow iS, e/\#$~~

$\downarrow i$

Chapter 7.

★ 综合属性

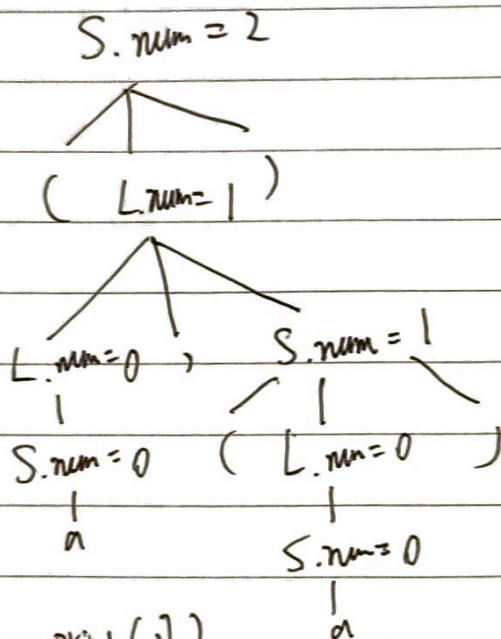
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1. $S' \rightarrow S \{ \text{Print}(S.val) \} \{ S'.val := S.val; \text{print}(S'.val); \}$
 $S \rightarrow S'VT \quad ; \quad \dots$
 $S \rightarrow T \quad ; \quad \dots$
 $T \rightarrow T' \wedge F \quad ; \quad \{ T.val := T'.val \text{ AND } F.val \}$
 $T \rightarrow F \quad \{ T.val := F.val \} \quad \dots$
 $F \rightarrow \neg F' \quad \{ F.val := \neg F'.val \} \quad \{ F.val := \text{NOT } F'.val \}$
 $F \rightarrow f \quad \{ F.val := f.lexval \} \quad \{ F.val := \text{false}; \}$
 $F \rightarrow t \quad \{ F.val := t.lexval \} \quad \{ F.val := \text{true}; \}$

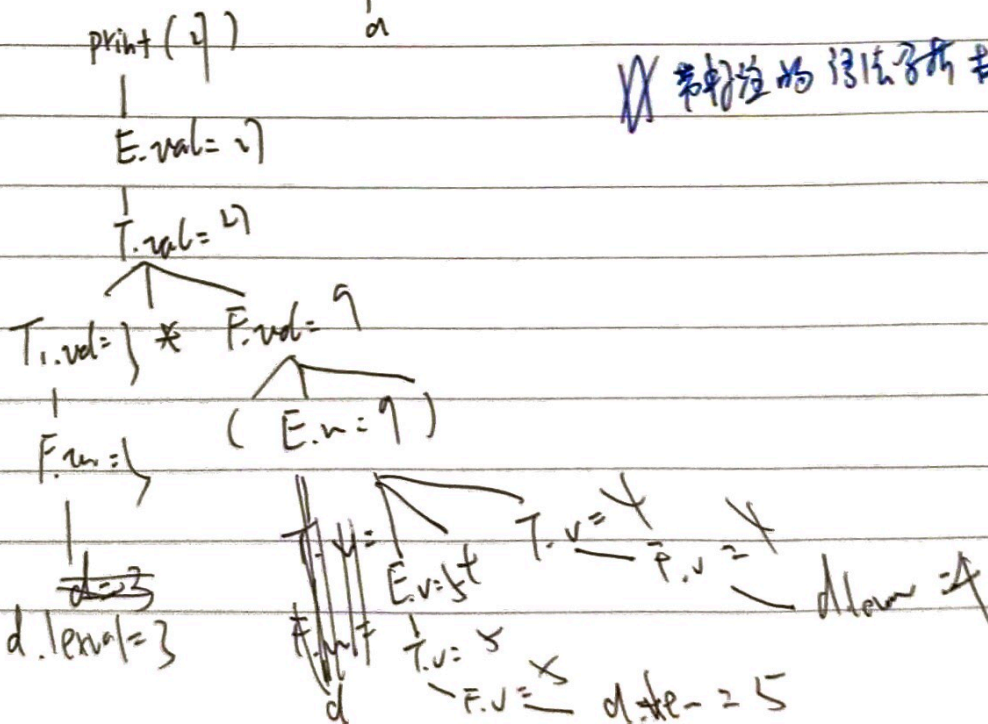
同前例上的赋值过程. 引入综合属性 $\rightarrow$ 语义属性.

2.



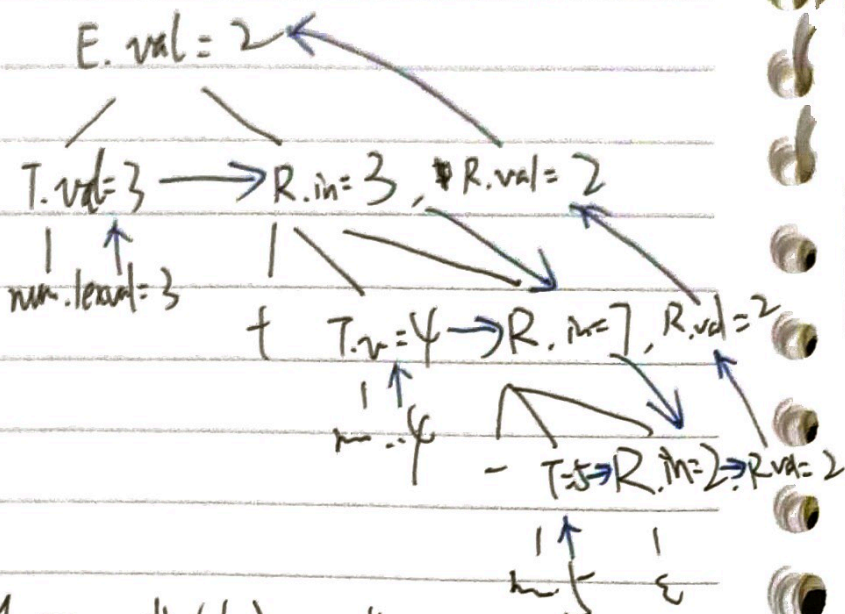
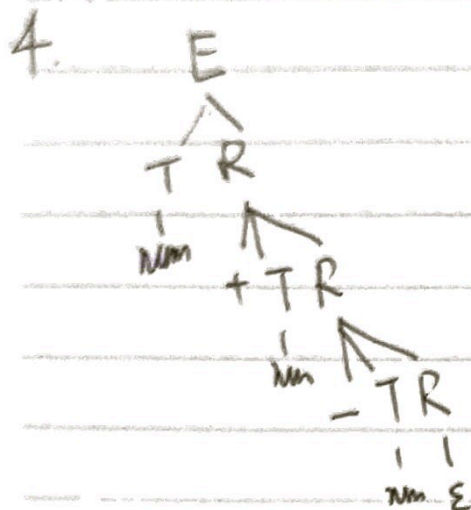
★ 综合 Tree.

3.



★ 翻译为语法树. (综合)

* L.



5. 14, 0246 -- 1 - # (L) #
15, 01 - 2 # S # * LR.

* * 8. $S \rightarrow E$. $\{ \text{print}(v[\text{top}].\text{val}); \}$ * LR + S.

$E \rightarrow E + T$. $\{ v[\text{top}+2].\text{val} = v[\text{top}+2].\text{val} + v[\text{top}].\text{val}; \}$

$E \rightarrow T$. $\{ v[\text{top}].\text{val} = v[\text{top}].\text{val}; \}$

...

$F \rightarrow (E)$. $\{ v[\text{top}+2].\text{val} = v[\text{top}+1].\text{val}; \}$

$F \rightarrow a$. $\{ v[\text{top}].\text{val} = \text{lexval}; \}$

9. $S \rightarrow MA_1B$

if $w[t] = 0$ ~~then~~ $v[t] = \text{true}$;
else ~~if~~ $v[t-1] = \text{false}$;

$A \rightarrow Aa$

$v[t+1] = w[t+1] - 1$

$A \rightarrow \epsilon$

$w[t+1] = w[t]$

* LR + L.

$B \rightarrow Ba$

$v[t+1] = v[t-1] - 1$

$B \rightarrow Bb$

$w[t+1] = w[t-1]$

$B \rightarrow \epsilon$

$w[t+1] = w[t-1]$

$M \rightarrow \epsilon$

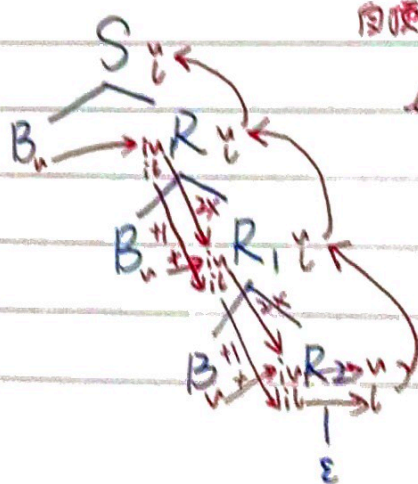
~~and~~ $w[t+1] = 100$

引入堆内存的表达式

10. $D \rightarrow D_1; T$ $M_1 \{ L.type := M_1.type; L.offset := M_1.offset; L.width = M_1.width; L \{ D.width := D_1.width + M_1.len \times T.width \}$
 $M_1 \rightarrow \epsilon \{ M_1.type := T.type; M_1.offset := D_1.width; \dots \}$

11. $N \rightarrow M_1 S \{ print(S.val) \}$ (2) $print(v[t].v)$
 $M_1 \rightarrow \epsilon \{ M_1.f := 1 \}$ $v[t+1].f = 1$
 $S \rightarrow M_2 S_1 M_3 B \{ S.v := S_1.v + B.v \}$ $v[t-1].v = v[t-2].v + v[t].v$
 $M_2 \rightarrow \epsilon \{ M_2.f := 2 \times S.f \}$ $v[t+1].f = 2 \times v[t].f$
 $M_3 \rightarrow \epsilon \{ M_3.f := S.f \}$ $v[t+1].f = v[t-2].f$
 $S \rightarrow \epsilon \{ S.v := 0 \}$ $v[t+1].v = 0$
 $B \rightarrow 0 \{ B.v := 0 \}$ $v[t].v = 0$
 $B \rightarrow 1 \{ B.v := B.f \}$ $v[t].v = v[t-1].f$

13. (1). $N \rightarrow S_1 S_2 \{ N.val := S_1.val + 2^{-S_2.len} \times S_2.val \}$
 消除左递归后
 引入 VN R 并构造
 设计规则和
 符号属性。
 $S \rightarrow B \{ R.in.val := B.val; R.in.len := 1 \}$ $R \{ S.val := R.val; S.len := R.len \}$
 $R \rightarrow B \{ R_1.in.val := 2 \times R.in.val + B.val; R_1.in.len = R.in.len + 1 \}$ $R_1 \{ R.val := R_1.val; R.len := R_1.len \}$
 $R \rightarrow \epsilon \{ R.val := R.in.val; R.len := R.in.len \}$
 $B \rightarrow 0 \{ B.val := 0 \}$
 $B \rightarrow 1$...
 接收地址
 作为累加值
 将结果送回顶部

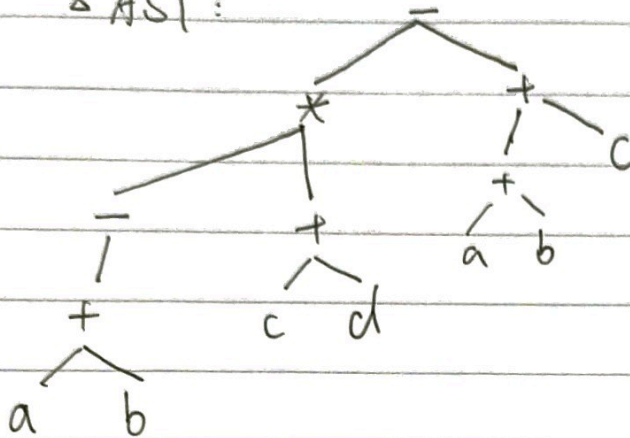


自顶向下翻译 (递归下降分析)

like 函数调用:
 B 调用函数 R,
 R 返回调用 R₁,
 最后向上传 result 值.

习题 1: $-(a+b) * (c+d) - (a+b+c)$ 表示成逆波兰式、AST 和四元式形式

Δ AST:



• 四元式: $(+, a, b, t_1)$

$$(-, t_1, -, t_2)$$
$$(t, c, d, t_3)$$
$$(*, t_2, t_3, t_4)$$
$$(+, a, b, t_5)$$
$$(t, t_s, c, t_b)$$
$$(c, t_4, t_6, t_7)$$

2. while ($A < B$) do

if $(C > D)$ and $E < F$ then

$$X := Y + Z$$

else

$$X := Y - Z$$

⑩ 3式:

```
while { 100 (j < , A, B, 102)
       101 (j , - , - , 112)
       102 (j > , C, D, 104)
       103 (j , - , - , 108)
       104 (j < , E, F, 106)
       105 (j , - , - , 108)
then { 106 (+, Y, Z, X) < 107
       108 (j , - , - , 100) (: = , t1 , - , X)
else { 109 (-, Y, Z, X) < 110
       111 (j , - , - , 100) (: = , t2 , - , X)
       112
```